

Hassaan Ali and Shyamal Gupta's EEC Research Proposal: Networking/Streaming

Executive Summary

Statement: Measure how networking “delivery choices” (client-server full download vs adaptive streaming vs peer-to-peer sending) and security/protocol choices (TLS vs plain, HTTP/1.1 vs 2 vs 3/QUIC, WideVine or PlayReady DRM vs non-encrypted, different containers like AV1, MKV, MP4) affect energy needed by both the server and the client to engage over the network with varying amounts of bandwidth.

Relation: It fits energy-efficient computing by focusing on both cloud computing (common industry server tasks needed) and communications systems (also touching on paradigms made standard through the internet). It explicitly focuses on the software/protocol decisions made which affect energy outcomes, and any leeway on making it more efficient.

Type: Project report that details something we designed, carried out, and then measured. We want to experiment as much as possible with the tools we have, and come to a consensus/findings on how this affects major players in networking tasks like Meta, YouTube, and Netflix (streaming at high bitrate with need for security but also efficiency).

Scope and Relevance

This idea from the get go was very appealing to us. As video content takes the world by storm, it's clear that there is a need for high bandwidth and good content delivery without wasting too much energy and resources but also delivering stability across a wide variety of users. Although we wanted the paper to be more overarching, it would make more sense for us to focus on video as we don't want to conquer the world of CDNs in one paper. We were aware of a project called Puffer (<https://puffer.stanford.edu/>), and that group at Stanford mainly focused on good algorithm choices that create stability over the network. We want to look at how choices made server-side for security and stability (like the ones used in the industry today and more experimental ones like Puffer) can affect how much energy is needed to keep up with today's rapid entertainment and video demand.

It is extremely relevant to this class because these distributed systems and server configurations take up large amounts of power. Although it doesn't compare to AI Data Center applications recently, these server configurations are going nowhere in the future. We've seen some preliminary papers that show adding TLS adds extra behavior but we want to estimate how much it makes a difference. Additionally, different HTTP protocol versions can also affect energy-constrained devices (like mobile/battery). In distributed delivery, things like P2P have been shown to help with location variety, but can be inconsistent. We want to show what choices are sound and energy efficient, while others may be consuming much more energy/resources for the sake of privacy.

In the paper, we want to implement a controlled testbed with an Ubuntu server and Linux/Mac client. We will try running three delivery modes: bulk file downloading over HTTP, adaptive bitrate video streaming (HLS/DASH as used on live TV solutions), and P2P file distribution. For each mode, we will sweep through the protocol/security configurations like TLS (on/off when applicable), HTTP Versions, and encryption (when possible). From this, we will report joules/MB delivered, joules per request (for chunked streams), and QoS metrics that are a must for both the client and server. We will focus on identifying when security and more advanced protocols reduce energy by shortening transfers versus when they increase energy by keeping CPU/network interfaces busy longer. These results will be displayed in a comprehensive paper, and we believe that this project should be feasible, given that the majority is testing software which we can do on the lab machines with a variety of tools/emulators and files to test on are a dime a dozen nowadays.

Preliminary References

- 1) TLS and energy consumption on a mobile device (<https://dl.acm.org/doi/10.1109/ISCC.2011.5983970>): showcasing how transmission/I/O energy can exceed computational energy for the actual cryptographic algorithms
- 2) Energy consumption of smartphones and IoT devices when using different versions of the HTTP protocol (<https://www.sciencedirect.com/science/article/pii/S1574119223001293>): comparing HTTP protocols across different communication patterns
- 3) BitTorrent on mobile phones - Energy efficiency of a distributed proxy solution (<https://cris.vtt.fi/en/publications/bittorrent-on-mobile-phones-energy-efficiency-of-a-distributed-pr/>): P2P energy efficiency studied, involving the role of proxying/caching, central server vs an entire peer to peer network of hosts